AP CALCULUS AB

Unit 1 • Lesson 1.14 practice

Lesson 1.14 • Connecting Infinite Limits and Vertical Asymptotes

4 multiple-choice questions and 1 free-response question

Question 1

$\lim_{x \to 0^-} \frac{1}{x}$ is

(A) $+\infty$

(B) $-\infty$

(C) 0

(D) 1

Question 2

$\lim_{x \to 2} \frac{3}{(x-2)^2}$ is

(A) $+\infty$

(B) $-\infty$

(C) 3

(D) It does not exist and is not infinite.

Question 3

Which line is a vertical asymptote of $f(x) = \ln(x-1)$?

(A) $x = 0$

(B) $x = e$

(C) $x = 1$

(D) $y = 1$

Question 4

$\lim_{x \to \frac{\pi}{2}^+} \tan x$ is

(A) $+\infty$

(B) 0

(C) 1

(D) $-\infty$

Question 5 (Free response)

Let $f(x) = \dfrac{x+2}{x^2-4x}$.

(a) Find all vertical asymptotes of the graph of f.

(b) For each asymptote, find both one-sided limits. Justify the signs.